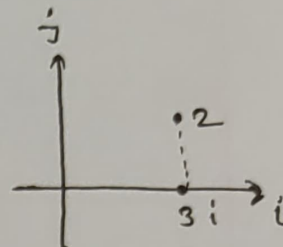


Waves and Oscillation:

Periodic Motion: Clock

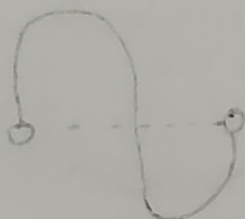
① Basis calculation of vector Analysis

② Unit vector system (m)



$\vec{x}$

$$v = \frac{dx}{dt}$$



speed $\cdot v = \frac{dx}{dt} \hat{i}$

" speed up $\uparrow$ acceleration
speed down $\downarrow$ de acceleration

acceleration $a = \frac{dv}{dt}$

$$= \frac{d}{dt} \cdot \left(\frac{dx}{dt} \right)$$

$$= \frac{d^2x}{dt^2}$$

Question

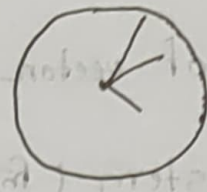
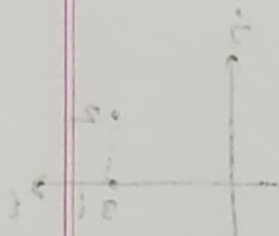
① Short Question - 6

② Math - 18

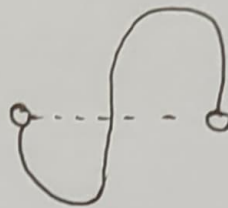
③ Proof -

Periodic Motion : Clock

Simple Harmonic Motion

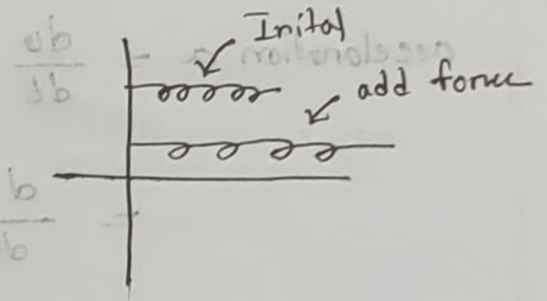
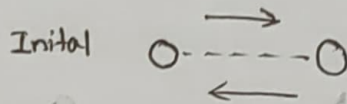


Oscillatory motion :



Simple Harmonic motion :

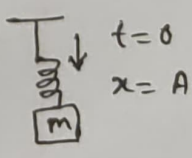
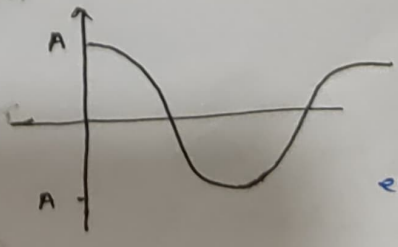
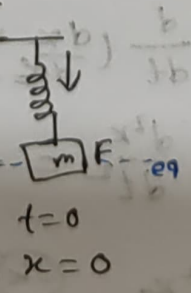
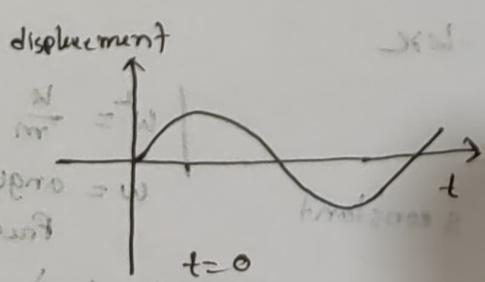
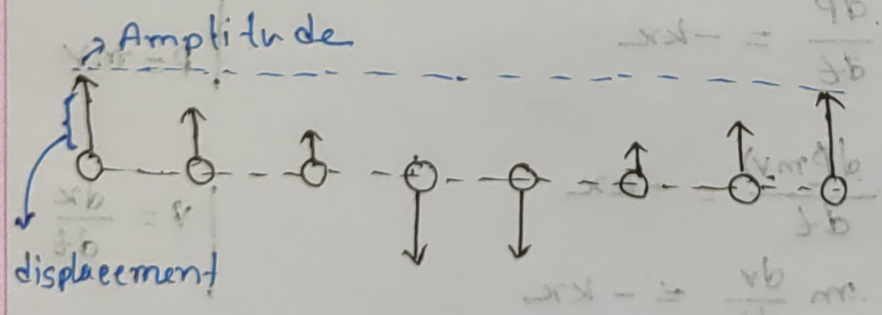
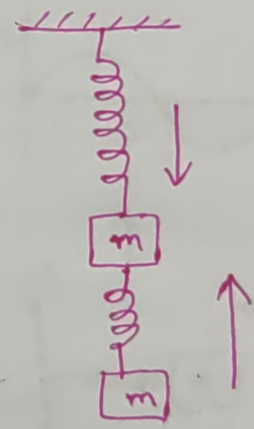
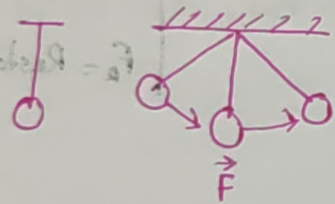
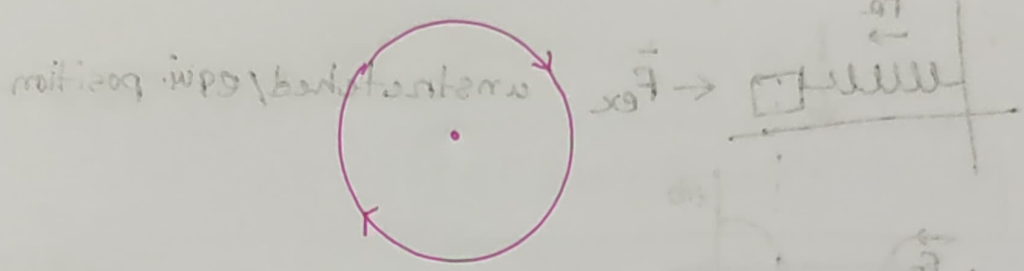
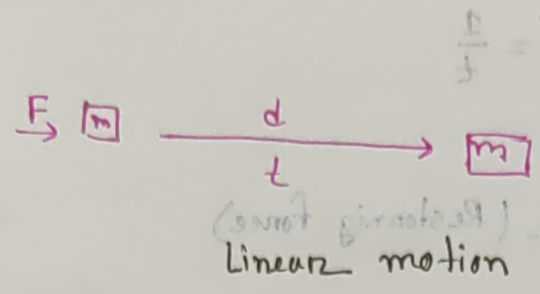
restoring force



Damped Harmonic motion :

SHM is a type of motion.

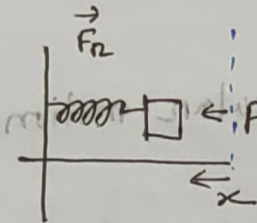
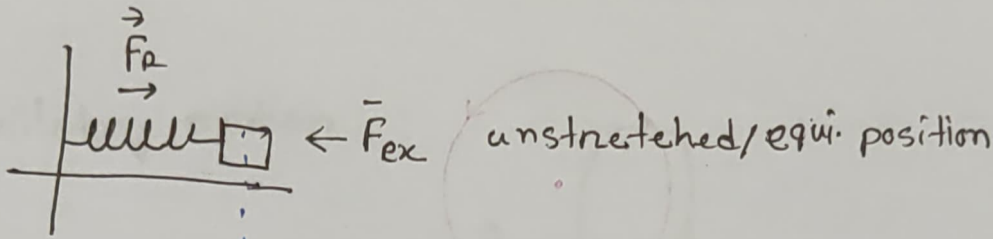
Different kind of motion:



Frequency: number of cycle in a unit of time

$$f = \frac{1}{T}$$

Mass Spring Oscillator (Restoring Force)



$$\vec{F}_R \propto -\vec{x}$$

$$\vec{F}_R = -k\vec{x}$$

k = spring constant
 F_R = Restoring Force

$$d\vec{p} = \vec{F} = -k\vec{x}$$

$$\frac{d\vec{p}}{dt} = -k\vec{x}$$

$$\frac{d(\vec{p} = m\vec{v})}{dt} = -k\vec{x}$$

$$m \frac{d\vec{v}}{dt} = -k\vec{x}$$

$$m \frac{d}{dt} \left(\frac{d\vec{x}}{dt} \right) = -k\vec{x}$$

$$m \frac{d^2\vec{x}}{dt^2} = -k\vec{x}$$

$$\omega^2 = \frac{k}{m}$$

ω = angular Frequency

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$\frac{d^2x}{dt^2} = \frac{-k}{m} x$ $\frac{d^2x}{dt^2} = -\omega^2 x$

$\omega^2 = \frac{k}{m}$ ω = angular Frequency

$x = \square +$ after x is value after change,

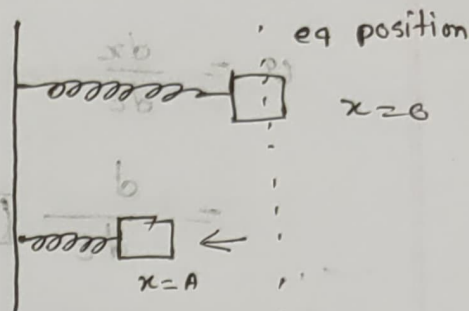
V_2

$a =$

$$f = m\omega$$

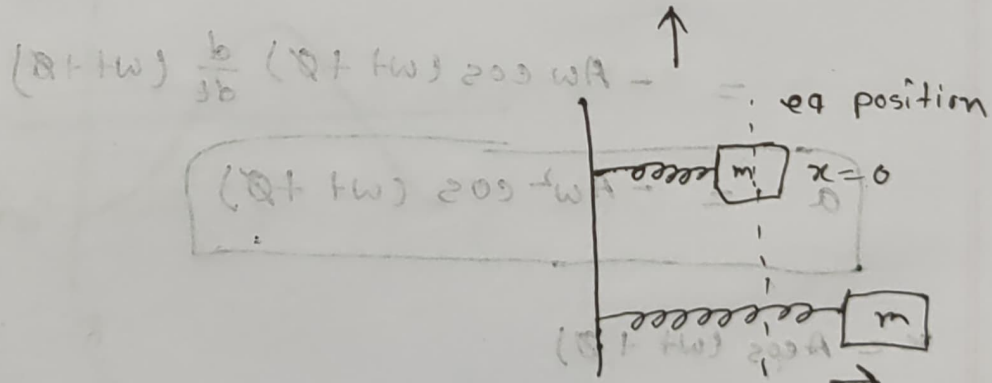
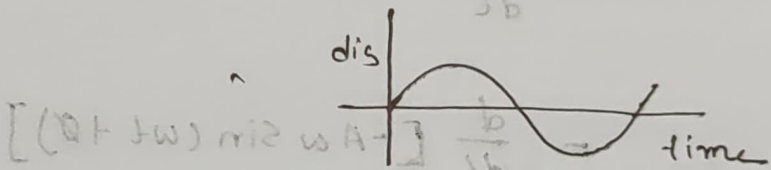
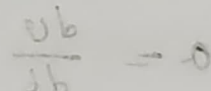
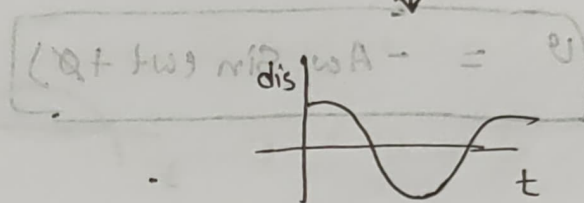
$$E_k = \frac{1}{2}mv^2$$

$$E_p = \frac{1}{2} k x^2$$



$$x = A \cos(\omega t + \phi)$$

$$x = A \sin (\omega t + f)$$



0-6 Hi

$$x = A \cos(\omega t + \phi)$$

$$v = \frac{dx}{dt}$$

$$= \frac{d}{dt} [A \cos(\omega t + \phi)]$$

$$= -A \sin(\omega t + \phi) \cdot \frac{d}{dt} (\omega t + \phi)$$

$$v = -A\omega \sin(\omega t + \phi)$$

$$a = \frac{dv}{dt}$$

$$= \frac{d}{dt} [-A\omega \sin(\omega t + \phi)]$$

$$= -A\omega \cos(\omega t + \phi) \cdot \frac{d}{dt} (\omega t + \phi)$$

$$a = -A\omega^2 \cos(\omega t + \phi)$$

$$x = A \cos(\omega t + \phi)$$

$$v = -A\omega \sin(\omega t + \phi)$$

$$a = -A\omega^2 \cos(\omega t + \phi)$$

$$\text{if } \phi = 0$$

$$x = A \cos \omega t$$

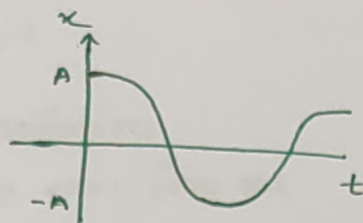
$$v = -A \sin \omega t$$

$$a = -A\omega^2 \cos \omega t$$

$$y = A \cos t$$

$$\begin{aligned} \frac{dx}{dt} &= A \sin(\omega t + \phi) \omega \\ A \cos(\omega t + \phi) \omega \\ A \sin(\omega t + \phi) \omega \end{aligned}$$

$$x = A \cos \omega t$$

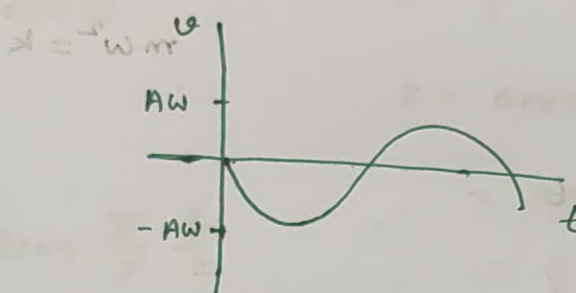


$$E = \frac{1}{2} k x^2$$

$$E = \frac{1}{2} k (A \cos \omega t)^2$$

$$E = \frac{1}{2} k A^2 \cos^2 \omega t$$

$$v = -A\omega \sin \omega t$$



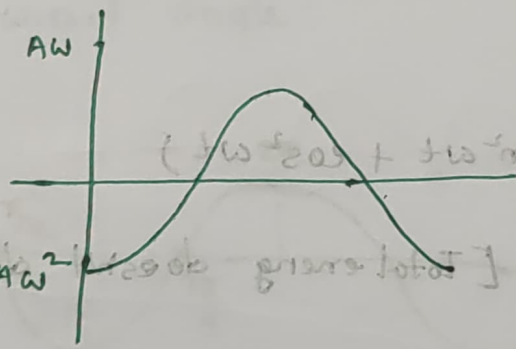
$A\omega = \text{velocity max}$

$$E = \frac{1}{2} m A^2 \omega^2 \sin^2 \omega t$$

$$E = \frac{1}{2} m A^2 \omega^2 \cos^2 \omega t$$

$$E = \frac{1}{2} m A^2 \omega^2$$

$$a = -A\omega^2 \cos \omega t$$



$$E = E_p + E_k$$

$$E = \frac{1}{2} k A^2 (\sin^2 \omega t + \cos^2 \omega t)$$

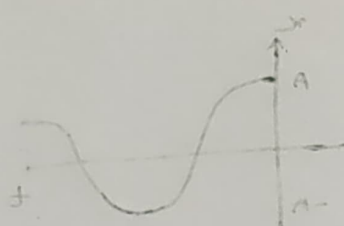
$$E = \frac{1}{2} k A^2$$

(Total energy is constant)

$$E_p = \frac{1}{2} k x^2$$

$$= \frac{1}{2} k (A \cos \omega t)^2$$

$$= \frac{1}{2} k A^2 \cos^2 \omega t$$



$$x = A \cos \omega t$$

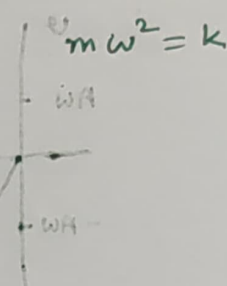
$$E_k = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m A^2 \omega^2 \sin^2 \omega t$$

$$= \frac{1}{2} m A^2 \frac{k}{m} \sin^2 \omega t$$

$$= \frac{1}{2} k A^2 \sin^2 \omega t$$

$$\omega^2 = \frac{k}{m} \Rightarrow \omega A = v$$



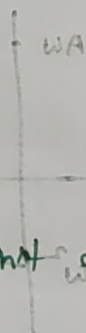
$$m \omega^2 = k$$

$$v = \omega A \sin \omega t$$

$$\therefore E_T = E_p + E_k$$

$$= \frac{1}{2} k A^2 (\sin^2 \omega t + \cos^2 \omega t)$$

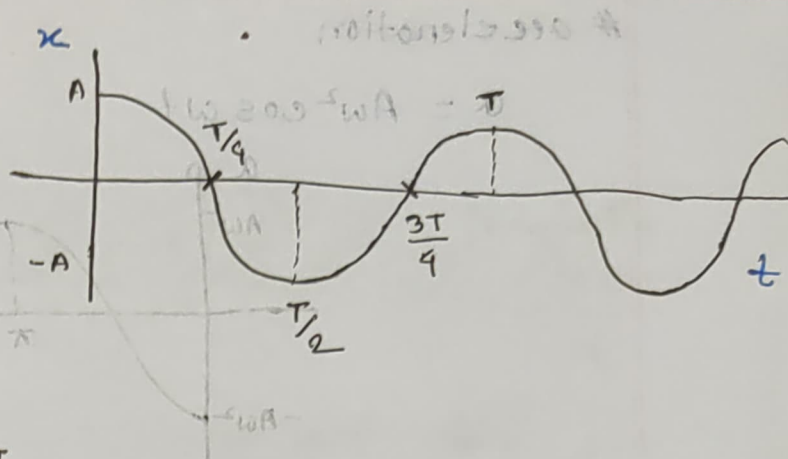
$$= \frac{1}{2} k A^2 \quad [\text{Total energy doesn't depend on time}]$$



$$x = A \cos \omega t$$

① total oscillation

② time to value $\frac{A}{2}$ ehk



① $t=0$

$$x = A \cos 0$$

$$= A$$

$$t = \frac{3T}{4}$$

$$x = A \cos \frac{2\pi}{T} \cdot \frac{3T}{4} = A \cos \frac{3\pi}{2} = -A$$

$$= A \cos \frac{3\pi}{2}$$

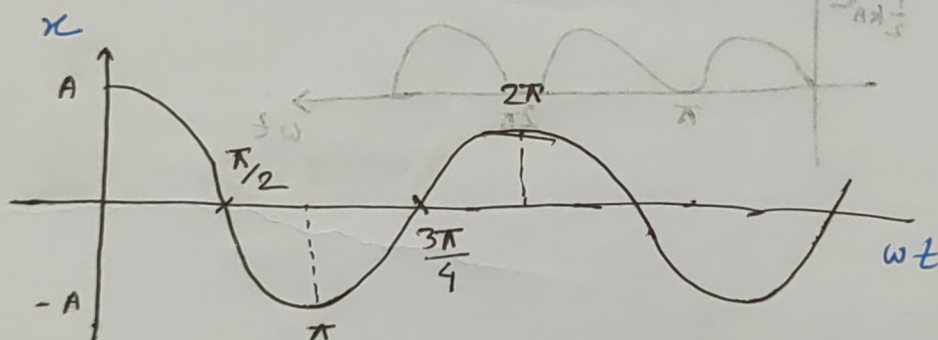
$$t = \frac{T}{2}$$

$$x = A \cos \frac{2\pi}{T} \cdot \frac{T}{2}$$

$$= A \cos \pi$$

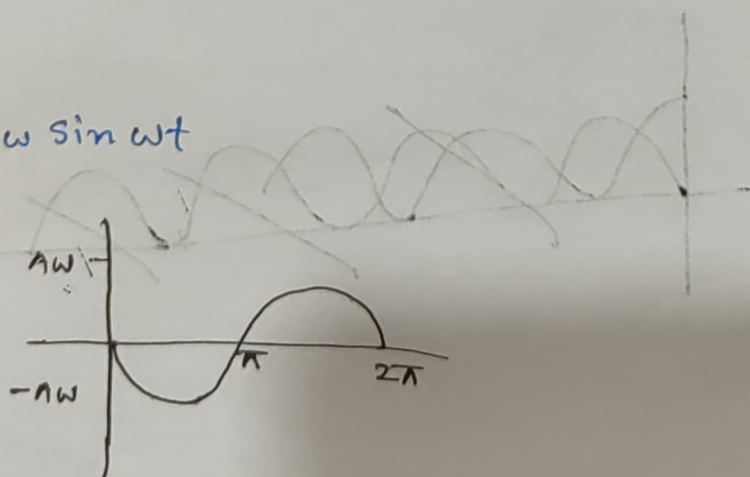
$$= -A$$

displacement w.r.t angle:



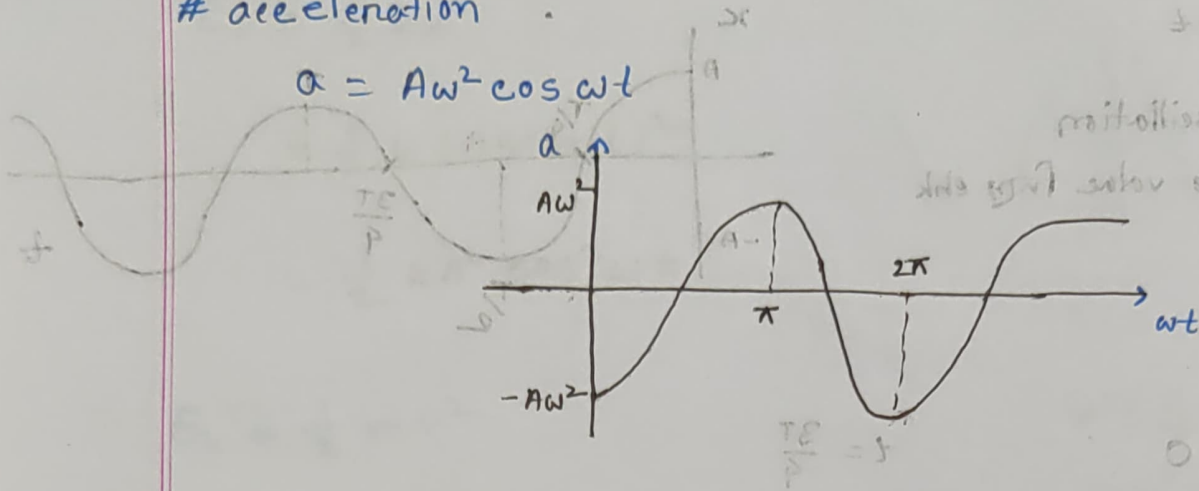
Velocity

$$v = -A\omega \sin \omega t$$



acceleration

$$a = A\omega^2 \cos \omega t$$



$$x = A \cos \omega t$$

mitteilung 1000 ①

das ist die selbe wie ②

0-1 ①

$$x = A \cos \omega t$$

$$A =$$

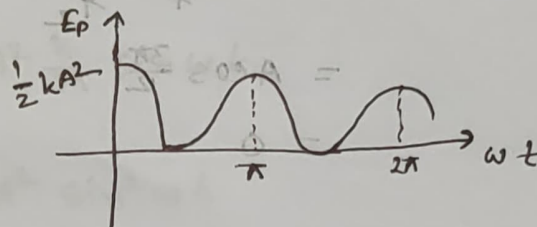
$$\frac{I}{\rho} =$$

$$\frac{1}{2} \frac{A^2}{\rho} \cos^2 \omega t =$$

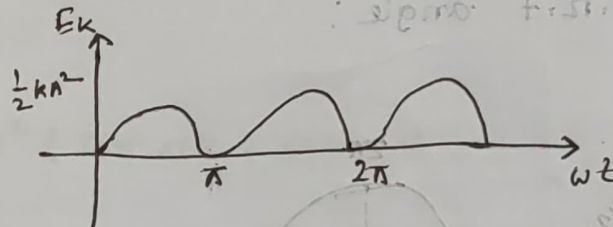
$$\frac{1}{2} \frac{A^2}{\rho} \cos^2 \omega t =$$

$$A =$$

$$\# E_p = \frac{1}{2} k A^2 \cos^2 \omega t$$

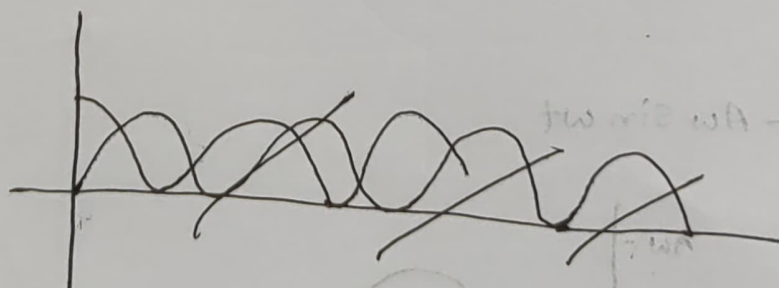


$$\# E_k = \frac{1}{2} k A^2 \sin^2 \omega t$$



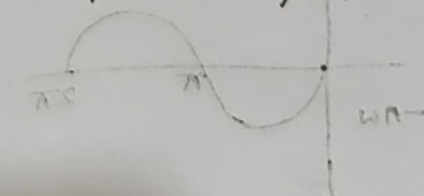
elasto f. s. w. trans. v. l. g. z. b.

E_T

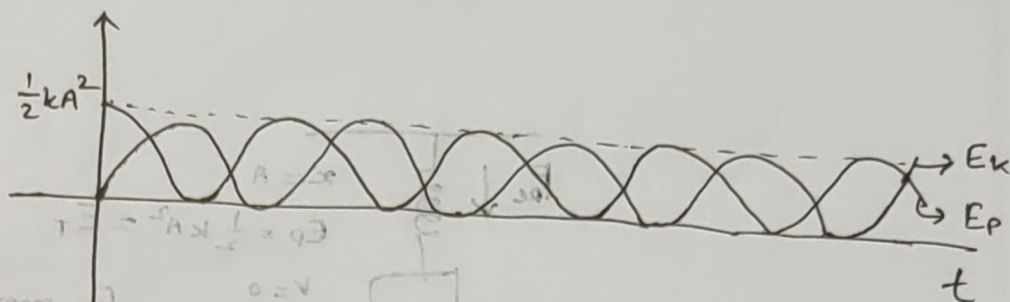


blissalav

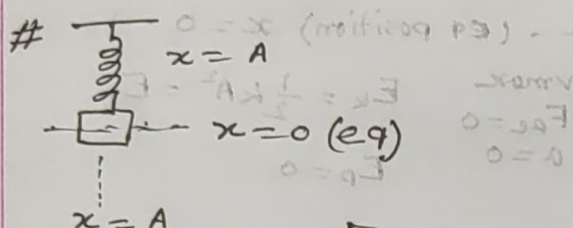
tw. m. s. w. f. s. w. v. l. g. z. b.



$$E_T =$$



all equation with value



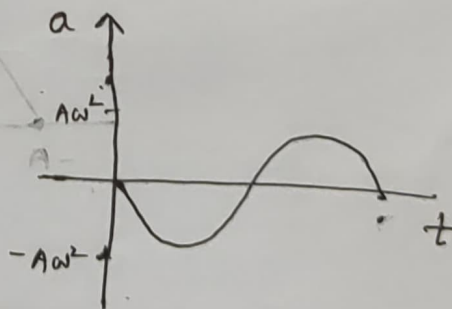
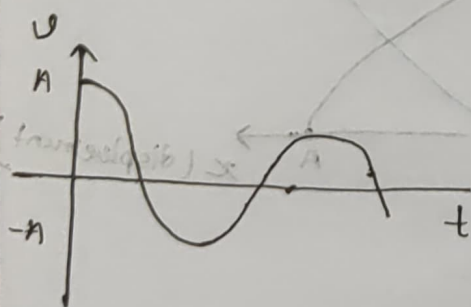
For this case
eq position

$$x = A \sin \omega t$$

$$v = \frac{dx}{dt}$$

$$= A\omega \cos \omega t$$

$$a = \frac{dv}{dt} = -A\omega^2 \sin \omega t$$



relation betⁿ a, v, E_p, E_k, E_T

$$a = \frac{v-u}{t}$$

• $x = A$

$$E_p = E_T = \frac{1}{2} k A^2$$

$$E_k = 0$$

• $x = 0$

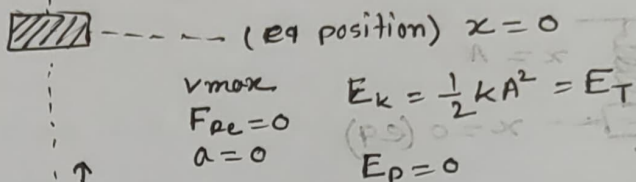
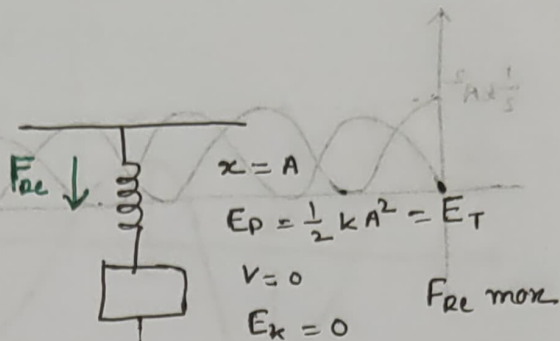
$$E_k = E_T = \frac{1}{2} A^2 k$$

$$E_p = 0$$

• $x = -A$

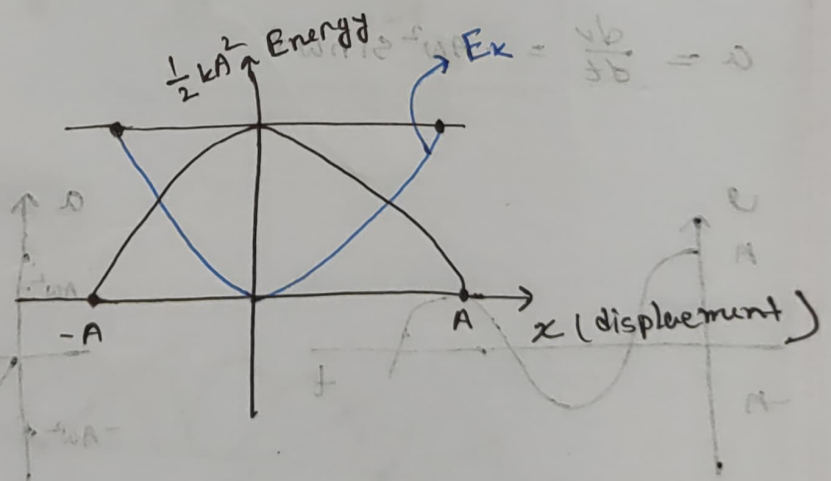
$$E_p = E_T = \frac{1}{2} k A^2$$

$$E_k = 0$$



$$E_p = \frac{1}{2} k A^2 = E_T ; E_k = 0$$

$x = -A$; Restoring (F_{spring}) max



L-4

03-10-2023

~~L-4~~
~~L-4~~

From part lecture....

Show that $v = \omega \sqrt{A^2 - x^2}$

$$E_T = E_p + E_k$$

$$\Rightarrow \frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$$

$$\Rightarrow \frac{1}{2} m v^2 = \frac{1}{2} k A^2 - \frac{1}{2} k x^2$$

$$\Rightarrow m v^2 = k A^2 - k x^2$$

$$\Rightarrow m v^2 = k (A^2 - x^2)$$

$$\Rightarrow v^2 = \frac{k}{m} (A^2 - x^2)$$

$$\Rightarrow v = \sqrt{\frac{k}{m} (A^2 - x^2)}$$

$$\Rightarrow v = \sqrt{\frac{k}{m}} \cdot \sqrt{A^2 - x^2}$$

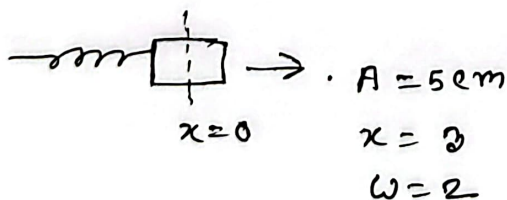
$$\Rightarrow v = \omega \sqrt{A^2 - x^2}$$

$$\left[\omega = \sqrt{\frac{k}{m}} \right]$$

$\therefore v = \omega \sqrt{A^2 - x^2} \rightarrow$ speed time of particle "

or, $v = -A\omega \sin \omega t \rightarrow$ speed time of particle.

Example:



$x=3$ Position $v = ?$

$$\begin{aligned} v &= \omega \sqrt{A^2 - x^2} \\ &= 2 \sqrt{5^2 - 3^2} \\ &= 8 \text{ cm s}^{-1} \end{aligned}$$

if given,
 $k = 19000$
 $m = 50 \text{ g}$

$$\begin{aligned} v &= \sqrt{\frac{k}{m}} \cdot \sqrt{A^2 - x^2} \\ &= \sqrt{\frac{19000}{50}} \cdot \sqrt{5^2 - 3^2} \\ &= 8\sqrt{95} \\ &= 77.97. \end{aligned}$$

if $x = 3 \cos 5t$ & $t = 2s$

$$v = \frac{dx}{dt}$$

$$= 3 \sin 5t \cdot 5$$

$$= 15 \sin 5t$$

$$= 15$$

Phase / फेज

Phase difference

$$\left(\frac{\pi}{p} + w\right) \text{ or } A = s$$

अन्य स्थिति में
अपेक्षित

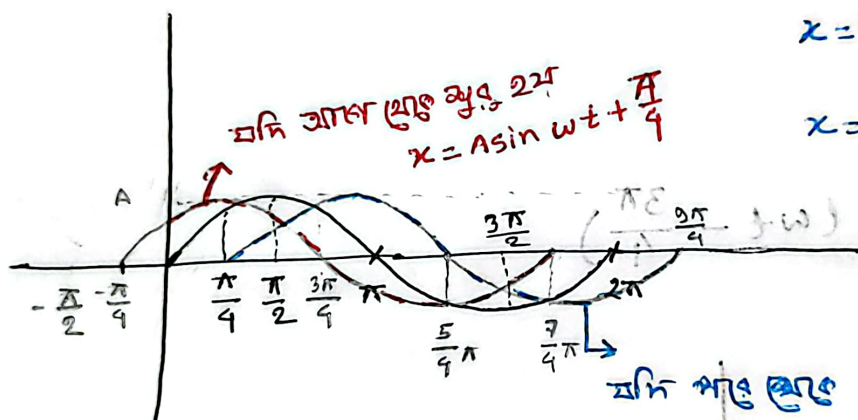
① is used to describe the phase position of one wave relative to another

$$x = A \sin \omega t$$

shifting the original graph w.r.t. to ωt using phase $[\theta]$

ज-व

displacement vs Angle graph.



$$x = A \sin \omega t + \theta \rightarrow$$

$$x = A \sin \omega t - \theta$$

middle point जहाँ $\frac{1st + last}{2}$

अन्य स्थिति में

अपेक्षित

$$x = A \sin \omega t - \frac{\pi}{4}$$

way 2

① value $\tan^{-1} 0$ जहाँ जहाँ जहाँ

② $\omega t + x$ or $\omega t - x$ जहाँ angle जहाँ जहाँ

③ '+' जहाँ left

④ '-' जहाँ Right

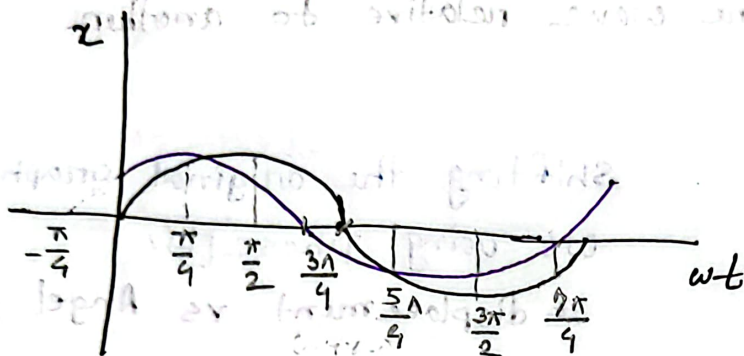
Question

① $x = A \sin(\omega t + \frac{\pi}{4})$

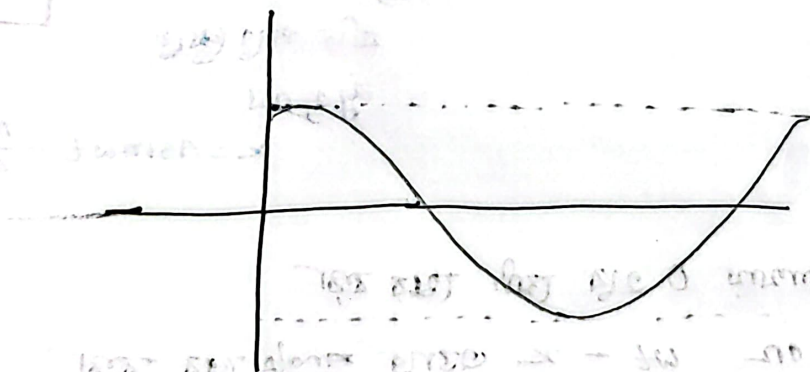
$\omega t + \frac{\pi}{4} = 0$

$\omega t = -\frac{\pi}{4}$

Construct new graph at



② $x = A \cos(\omega t - \frac{3\pi}{4})$

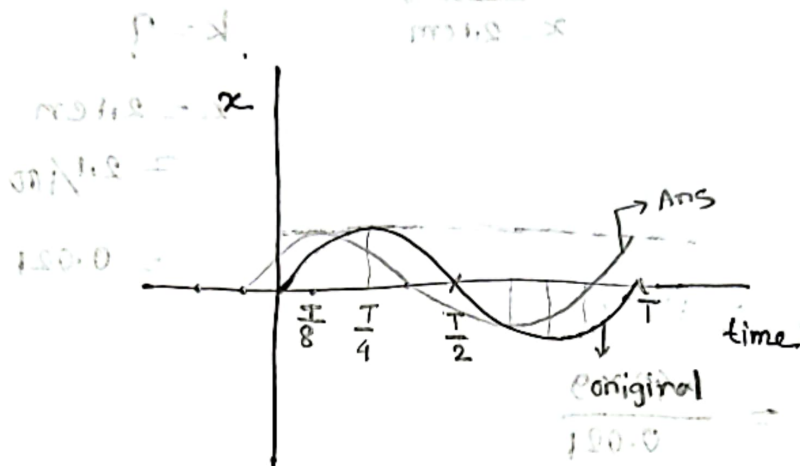


Let $x = A \sin(\omega t + \frac{\pi}{4})$ (iii)

Let $x = A \cos(\omega t - \frac{3\pi}{4})$ (iv)

Time

$$x = A \sin(\omega t + \frac{\pi}{4})$$



$$\omega t = -\frac{\pi}{4}$$

$$\frac{2\pi}{T} t = -\frac{\pi}{4}$$

$$t = -\frac{\pi}{4} \times \frac{T}{2\pi}$$

$$= -\frac{T\pi}{8\pi}$$

$$= -\frac{T}{8}$$

Start here

$$x = A \sin(\omega t - \frac{\pi}{2})$$

$$k = 10.6 \text{ N/m}$$

$$A = 0.100 \text{ m}$$

$$x = 0.100 \sin(8.08t)$$

$$v = 0.808 \sin(8.08t)$$

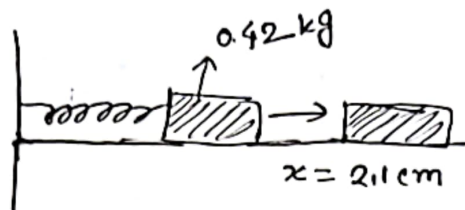
$$(x, v, a)$$

$$v = 0.808 \sin(8.08t)$$

$$v = 0.808 \sin(8.08t)$$

Meth.

①



$$m = 0.42 \text{ kg}$$

$$a = 9.0 \text{ ms}^{-2}$$

$$k = ?$$

$$x = 2.1 \text{ cm}$$

$$= 2.1/100$$

$$= 0.021$$

$$F = kx$$

$$k = \frac{F}{x}$$

$$= \frac{ma}{x} = \frac{0.42 \times 9}{0.021}$$

$$= 180 \text{ N/m}$$

②

$$k = 19.6 \text{ N/m}$$

$$A = 0.100 \text{ m}$$

$$x = -0.1 \cos 8.08t$$

$$v = 0.808 \sin 8.08t$$

a) total energy?

b) E_k and E_p as a function of time

c) the velocity when the mass is 0.050 m from eqb

d) E_k and E_p at half Amplitude ($x = \pm A/2$)Soln

$$a) E_T = \frac{1}{2} k A^2$$

$$= \frac{1}{2} \times 19.6 \times (0.100)^2$$

$$\textcircled{b} E_k = \frac{1}{2} m v^2$$

$$= \frac{1}{2} \times m (0.008 \sin 8.08t)^2$$

$$= \frac{1}{2} m (0.008)^2 \sin^2 8.08t$$

$$E_p = \frac{1}{2} k x^2$$

$$= \frac{1}{2} \times 19.6 \times (-0.1 \cos 8.08t)^2$$

$$= \frac{1}{2} \times 19.6 \times (0.1)^2 \cos^2 8.08t$$

$$\textcircled{c} A = 0.1 \text{ m}$$

$$x = 0.05 \text{ m}$$

$$x = -0.1 \cos 8.08t$$

$$\therefore \omega = 8.08$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

$$= 8.08 \sqrt{(0.1)^2 - (0.05)^2}$$

$$= 0.60$$

$$E_k + E_p = \frac{1}{2} k x^2$$

$$= \frac{1}{2} k \left(\frac{A}{2}\right)^2$$

$$= \frac{1}{2} k \cdot \frac{A^2}{4}$$

$$= \frac{1}{4} \cdot \frac{1}{2} k A^2$$

$$= \frac{1}{4} E_T$$

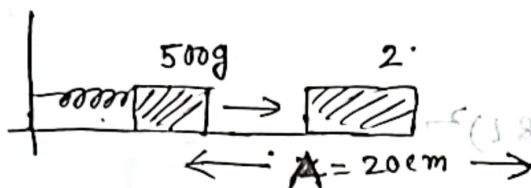
$$E_k = E_T - E_p$$

$$\therefore E_k = E_T - E_p$$

$$= E_T - \frac{1}{4} E_T = \frac{3}{4} E_T = \frac{3}{4} \cdot \frac{1}{2} k A^2 = \frac{3}{8} k A^2$$

$$= \frac{3}{8} E_T$$

③



$$A = 20 \text{ cm} = 0.2 \text{ m}$$

$$m = 500 \text{ g} = 0.5 \text{ kg}$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$T = 0.8 \text{ s}$$

$$\Rightarrow v = \frac{2\pi}{T} \sqrt{A^2 - x^2}$$

$$v = 1.0 \text{ m s}^{-1}$$

$$\Rightarrow \frac{v^2}{\left(\frac{2\pi}{T}\right)^2} = A^2 - x^2$$

$$\Rightarrow x^2 = A^2 - \frac{v^2}{\left(\frac{2\pi}{T}\right)^2}$$

$$\Rightarrow x = \pm \sqrt{A^2 - \frac{v^2}{\left(\frac{2\pi}{T}\right)^2}}$$

$$= \pm \sqrt{(0.2)^2 - \frac{(1)^2}{\left(\frac{2\pi}{0.8}\right)^2}}$$

$$x = \pm 1154 \cdot A \text{ mm}$$

④

$$x = A \sin \omega t$$

$$v = \frac{dx}{dt}$$

$$= A \cos \omega t \omega$$

$$= A \omega \cos \omega t$$

$$\therefore v_{\text{max}} = A \omega$$

$$= (0.2 \times 10^{-3}) \times 2\pi \times 1 \times 10^3$$

$$= 1.256 \text{ m s}^{-1}$$

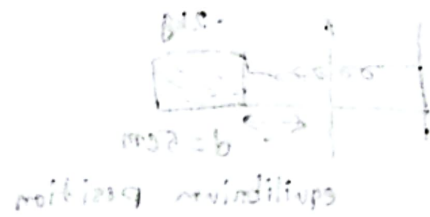
$$A = 0.2 \text{ mm}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$f = 1 \text{ kHz} = 1 \times 10^3 \text{ Hz}$$

⑥

100 MHz
 $\beta = 2 \times 10^8$



⑦

$$\omega = \sqrt{\frac{\pi}{\epsilon}} \cdot \sqrt{\frac{\epsilon}{\mu}} = \omega$$

$$\frac{\pi \epsilon}{\mu} = \omega^2 = \omega$$

$$\omega = \omega$$

$$\omega = \frac{\pi \epsilon}{\mu} \rightarrow \frac{\omega}{\pi \epsilon} = \frac{1}{\mu}$$

⑤ $f = 100 \text{ MHz}$
 $= 1 \times 10^8 \text{ Hz}$

$$\frac{\pi \epsilon}{\omega} = T$$

$T = ?$

$$\frac{\pi \epsilon}{\omega} = T$$

$$T = \frac{1}{f} = \frac{1}{1 \times 10^8}$$

$$= 1 \times 10^{-8} \text{ s}$$

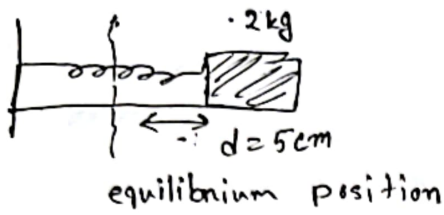
$$\omega = 2 \times 10^8 \text{ rad/s}$$

Amplitude and angular velocity of the wave

⑥ Amplitude and angular velocity of the wave

Amplitude
 II

7



$$k = 196 \text{ N/m}$$

$$m = 2 \text{ kg}$$

d

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 7\sqrt{2}$$

$$\omega = 2\pi f$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$f = \frac{\omega}{2\pi} = \frac{7\sqrt{2}}{2\pi} = 1.575$$

$$T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{7\sqrt{2}}$$

$$= 0.639$$

SHM 001

SH 001

C = T

$$\frac{1}{2\pi f} = \frac{1}{\omega} = T$$

$$2\pi f = \omega$$

b

$$x = 5 \cos 7\sqrt{2}t$$

released at $t=0$ s. so, initial Phase/0 means in

position at $t=0$ s. so, initial Phase/0 means in

SHM

8

a) 15 completed at $\cdot 10 \text{ s}$ / or $T = \frac{1}{f}$
 $1 \quad \quad \quad \frac{10}{15} \text{ s}$
 $= 0.667$

b)

$$V_{\max} = \omega A$$

$$A = 2 \text{ cm.}$$

$$= 0.2 \text{ m}$$

$$= \frac{2\pi}{T} \times 0.2$$

$$= 1.88 \text{ ms}^{-1}$$

c)

$$\omega = \frac{2\pi}{T} = 9.38$$

$$x = A \cos \omega t$$

$$= 0.2 \cos 9.38 t$$

$$v = -A\omega \sin \omega t$$

$$= -(0.2 \times 9.38) \sin 9.38 t$$

$$= -1.88 \sin 9.38 t$$

$$= -1.79 \text{ ms}^{-1}$$

$$t = 0.8 \text{ s}$$

calculator $\rightarrow$ Radian

mood $\rightarrow$ calculate

answer